Week 4 BA

2024 July07 Quiz-1

Comprehension for questions 1-2 (source Q14)

The linear demand response for product-A is modelled as a simple linear regression represented as D(P) = 2800 - 40*P, where D(P) is the demand at price-P. Then, answer the given sub questions

1. [Source Q15; SA]
What is the elasticity of the demand response curve when the price is Rs.50? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 2.50]

2. [Source Q16; SA]
What is the satiating price for the demand response curve? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 70.00]

2024 Oct27 Quiz-1

Comprehension for questions 3-5 (source Q12)

The linear demand response for product-A is modelled as a simple linear regression represented as D(P) = 4500 - 30*P, where D(P) is the demand at price-P. Then, answer the given subquestions.

3. [Source Q13; SA]
What is the elasticity of the demand response curve when the price is Rs.50? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 0.49 to 0.51]

4. [Source Q14; SA]
What is the satiating price for the demand response curve? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 149.99 to 150.99]

5. [Source Q15; SA]
What is the market size for the demand response curve? (Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 4500]

2024 Dec01 Quiz-2

6. [Source Q2; MCQ]
Latent demand in a demand-response curve is the area obtained when?

(a) Price is reduced below the identified optimal price

(b) Price is increased beyond the identified optimal price

(c) The optimal price is increased beyond the maximum available price

(d) Quantity is reduced below the identified optimal quantity

[Source ans: a]

2024 Dec22 ET

Comprehension for question 7 (source Q14)

A doctor is of the opinion that "Blood Pressure (BP)" is affected by "Age" and "Weight". Based on past data, the correlation between these three variables is identified and provided in Table-3. Then answer the given subquestions.

	BP	Age	Weight
BP	--		
Age	0.695	--	
Weight	0.950	0.507	--

Table-3

7. [Source Q15; SA]

If a linear regression model is built between "BP" and "Age", then how much percentage of variability in BP can be explained by Age? (Note: Enter your answer in percentage rounded to two decimal places. Eg: If your answer is "10.256%", then enter the answer as "10.26")

[Source ans: 48.30 to 48.32]

2025 Feb23 Quiz-1

Comprehension for questions 8-11 (source Q15)

(The following comprehension is purely imaginary) A local grosser has observed that the demand for Milk follows a linear demand response curve. Hence, he has decided to raise the "Milk" price by Rs. 5 rupees per litre to Rs. 45 per litre. If the demand was 300 litres per day before the rise and is now 100 litres per day after the rise, then answer the given subquestions.

8. [Source Q16; SA]
What is the elasticity of the linear demand response curve? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 40]

9. [Source Q17; SA]
What is the market size for milk? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 1900]

10. [Source Q18; SA]
What is the satiating price for the Milk's demand response curve? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 47.5]

11. [Source Q19; MSQ]
Given the computed elasticity, what can you say about the "Milk" demand? (choose all that is/are applicable)

(a) It is elastic

(b) It is inelastic

(c) It is a luxury

(d) It is a necessity

(e) Cannot say, insufficient information

[Source ans: a]

2025 Mar16 Quiz-2

12. [Source Q4; MSQ]
Latent demand in a demand-response curve is the area obtained when?

(a) Price is reduced below the identified optimal price

(b) Price is increased beyond the identified optimal price

(c) The optimal price is increased beyond the maximum available price

(d) Quantity is reduced below the identified optimal quantity

(e) Quantity is increased beyond the identified maximum quantity

[Source ans: a]

2025 Apr13 ET

Comprehension for questions 13-17 (source Q10)

(The following is a purely imaginary scenario) A demand response curve is modelled using linear regression. The partial regression output is given in Figure-1 below. Given this information, answer the subquestions.

ANOVA		
	df	*SS*
Regression		7076613
Residual		
Total	9	7184891
	Coefficients	*Standard Error*
Intercept	39942	
X Variable 1	-29.5	

Figure-1

13. [Source Q11; SA]
What is the elasticity of the demand response curve at a price of Rs. 50? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 0.02 to 0.05]

14. [Source Q12; SA]
What is the market size? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 39942]

15. [Source Q13; SA]
What is the satiating price? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

[Source ans: 1352 to 1355]

16. [Source Q14; MCQ]
Based on the elasticity, which of the following statements are TRUE

(a) The demand is elastic

(b) The demand is inelastic

(c) The price is elastic

(d) The price is inelastic

[Source ans: b]

17. [Source Q15; SA]
What percentage of the total linear variability in demand is captured by this model (given in figure-1)? (Note: Enter the answer in "percentage rounded to two decimal places without the percentage sign. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 98.00 to 99.00]

Comprehension for questions 18-19 (source Q16)

(The following is a purely imaginary scenario) A survey was conducted among 100 students who participated in all events at IITMs recent "Saarang" event. The students were asked to rate various aspects of their experience on a scale of "-5 to +5", where "-5" indicates a very negative experience and "+5" indicates a very positive experience. The average student rating for each event was taken across four key parameters: "Event Ambience", "Fairness of Event Judges", "Event Conduct", "Event Prize Money". The target variable was "Event Performance" which measures the overall performance of each event. Based on the collected data, a correlation matrix as specified in Table-2 is obtained for the independent variables. With this information, answer the given sub-questions

	Pairwise Correlation Values			
	Event Ambience	Fairness of Event Judges	Event Conduct	Event Prize Money
Event Ambience	1	-0.04	0.49	0.55
Fairness of Event Judges	-0.04	1	-0.011	0.022
Event Conduct	0.49	-0.011	1	-0.51
Event Prize Money	0.55	0.022	-0.51	1

Table-2

18. [Source Q17; SA]
What would be the R-Square value if a model is built with "Event Conduct" as the independent variable and "Event Prize Money" as the dependent variable? (Note: Enter the answer in percentage rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 25.00 to 27.00]

19. [Source Q18; SA]
What would be the Adjusted R-Square value if a model is built between "Fairness of Event Judges" as the independent variable and "Event Ambience" as the dependent variable? (Note: Enter the answer in percentage rounded to two decimal places without the "%" symbol. For example, if the answer is "1.234%", then enter it as "1.23")

[Source ans: 15.00 to 16.00]

2025 Aug03 Quiz-2

Comprehension for question 20 (source Q2)

Data on "Urban Population (UP)" of a country and the "GDP" of the same country are collected. The "GDP" is captured as a percentage (that is "10.5%" is captured as "10.5") and the "UP" is captured in crores of people (that is "1,20,00,000" is captured as "1.2"). From the raw data, the analyst has identified that the relation is non-linear. Hence, the analyst applies a natural logarithmic transformation on both variables (the independent variable is "GDP" and the dependent variable is "UP"). The transformed data is modelled as a Simple Linear Regression and the developed model is presented in Figure-1. Using this information, answer the given subquestions.

	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
Ln(GPD)	10.43004	.8278521	12.599	0.000	8.792235	12.06785
_cons	-24.42095	6.295892	-3.879	0.000	-36.87662	-11.96528

Figure-1

20. [Source Q3; SA]
How many degrees of freedom will be present for the "Regression" term in the ANOVA table (nomenclature as per excel)?

[Source ans: 1]

2025 Aug31 ET

Comprehension for questions 21-22 (source Q13)

The Following Comprehension and Related Subquestions are Purely Hypothetical Table-2 provides the Cumulative Television Rating Points (CTRP), money spend on promotion (in Lakhs, denoted by "P"), and the advertisement revenue (in Lakhs, denoted by "R") generated over a one-month period for 8 different television programs. The Revenue a TV show makes is the Dependent Variable. Using this data, answer the given subquestions.

TV Program ID	CTRP	P (in Lakhs)	R (in Lakhs)
TVP001	133	1.11	11.97
TVP002	111	1.04	10.53
TVP003	129	0.97	11.24
TVP004	117	0.79	9.87
TVP005	130	0.13	12.83
TVP006	154	1.08	12.95
TVP007	149	1.47	14.07
TVP008	90	1.04	9.22

Table-2

21. [Source Q14; SA]
What is the total variability in the dependent variable? [Note: Enter your answer in decimal rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: 18.00 to 21.00]

22. [Source Q15; SA]
The correlation coefficient between "CTRP" and "P" is 0.19. Then, what will be the Adjusted R-Square for the model (built using data in Table-2), where "CTRP" is the response variable and "P" is the predictor? [Note: Enter your answer in decimal rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: -0.14 to -0.12]

2025 Oct26 Quiz-1

23. [Source Q2; MCQ]
Latent demand in a demand-response curve is the area obtained when?

(a) Price is reduced below the identified optimal price

(b) Price is increased beyond the identified optimal price

(c) The optimal price is increased beyond the maximum available price

(d) Quantity is reduced below the identified optimal quantity

[Source ans: a]

Comprehension for questions 24-26 (source Q8)

(The following is a purely imaginary scenario) The demand for ice creams at different prices in the supermarket at IITM is specified in Table-1. Given this data, answer the subquestions (Note: For all the sub-questions, round your answer to two decimal places. Eg: If your answer is 10.256, then round it to 10.26)

Price	Ice-cream Demand
10	160
15	140
20	120
25	100
30	80

Table-1

24. [Source Q9; SA]
If a linear demand-response curve is fit for the data in Table-1, what is the satiating price for ice creams? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 50]

25. [Source Q10; SA]
If a linear demand-response curve is fit for the data in Table-1, what is the market size for ice creams? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 200]

26. [Source Q11; SA]
What is the elasticity of the linear demand-response curve, when the price is Rs. 35? (Note: Enter the answer rounded to two decimal places . For example, if the answer is "1.235", then enter it as "1.24")

[Source ans: 2.30 to 2.40]

2026 Mar15 Quiz-1

Comprehension for questions 27-31 (source Q2)

The demand response curve for a product is believed to be to linear. Therefore, the relationship between demand and price is modelled as a simple linear regression. The regression model built using 20 data points is as specified fellow. Demand = 5482 - 175*Price Based on the above data, answer the given subquestions.

27. [Source Q3; SA]
How many degrees of freedom is present for the "Residuals" of the built model?

[Source ans: 18]

28. [Source Q4; SA]
What is the market size?

[Source ans: 5482]

29. [Source Q5; SA]
What is the satiating price for the demand response curve? (Note: round your final answer to two decimal places. Example: If your answer is "1.2354" enter it as "1.24")

[Source ans: 31.00 to 31.50]

30. [Source Q6; SA]
What is the elasticity at a price of Rs. 20? (Note: round your final answer to two decimal places. Example: If your answer is "1.2354" enter it as "1.24")

[Source ans: 1.73 to 1.79]

31. [Source Q7; SA]
If the price changes from Rs.20 to Rs.25, then what is the elasticity of the curve between these two price points? (Note: round your final answer to two decimal places. Example: If your answer is "1.2354" enter it as "1.24")

[Source ans: 1.73 to 1.79]

2026 Apr06 Quiz-2

Comprehension for questions 32-36 (source Q1)

Milo Metals, a metal works shop, wants to predict the "Profit" for its products based on the expenditure spent on areas like "Quality", "Design" and "Delivery Time". Accordingly, data was collected for 50 different products it manufactured and sold. Milo Jr., the great grand son of Milo Sr., having joined the BS program, is interested to use regression as a tool to predict the profit. Accordingly, using the 50 sample points, he build the following FOUR models Model-1: Profit vs Quality Model-2: Profit vs Design Model-3: Profit vs Delivery Time Model-4: Profit vs Quality,Design,Delivery Time The partial regression outputs for the FOUR models are provided in Figures 1 to 4. Additionally, Table-1 provides the correlation matrix for the various columns in the data set.

Regression Statistics	
Multiple R	
R Square	
Adjusted R Square	0.95
Standard Error	9.4164
Observations	

Figure-1: Partial Regression Output for Model-1

Using this information, answer the given sub-questions

ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression		**P1**		**P2**	
Residual		76397.77			
Total					

Figure-2: Partial Regression Output for Model-2

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	90.07	8.239		1.26E-14
Delivery Time	0.25	0.032	**P3**	4.38E-10

Figure-3: Partial Regression Output for Model-3

ANOVA				
	df	*SS*	*MS*	*F*
Regression	**P4**	75683.9642		
Residual	**P5**			
Total				

Figure-4: Partial Regression Output for Model-4

	Profit	*Quality*	*Design*	*Delivery Time*
Profit	1			
Quality	0.97	1		
Design	0.2	0.24	1	
Delivery Time	0.75	0.72	-0.03	1

Table-1: Correlation Matrix

32. [Source Q2; SA]
What is the total variability present in error terms for Model-1? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 4250.00 to 4261.00]

33. [Source Q3; SA]
What is the total variability present in dependent variable? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 79600.00 to 79610.00]

34. [Source Q4; SA]
What is the value for P1? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 3200.00 to 3210.00]

35. [Source Q5; SA]
What is the value for P2? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 1.8 to 2.2]

36. [Source Q6; SA]
What is the value for P3? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 7.70 to 7.90]

2026 May06 ET

37. [Source Q29; MCQ]
What is the satiating price for a linear demand response curve?

(a) Zero

(b) Does not exist

(c) Equal to the Market size

(d) None of these

[Source ans: d]

Comprehension for questions 38-40 (source Q30)

The following scenario is purely hypothetical The demand for ice creams at different prices in the supermarket at IITM is specified in Table-2. Given this data, answer the given subquestions

Price	Ice-cream Demand
100	600
150	800
200	1000
250	1200
300	1400

Table-2

38. [Source Q31; SA]
If a linear demand-response curve is fit for the data, what is the satiating price for ice creams? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 374.00 to 376.00]

39. [Source Q32; SA]
If a linear demand-response curve is fit for the data, what is the market size for ice creams? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 1499.00 to 1501.00]

40. [Source Q33; SA]
What is the elasticity of the linear demand-response curve, when the price is Rs. 175? (Note: Enter your answer in decimal values rounded to two decimal places. For example, if your answer is "1.235", then enter the answer as "1.24")

[Source ans: 0.85 to 0.89]

2026 Jul16 Quiz-1

Comprehension for questions 41-44 (source Q1)

(The following comprehension is purely imaginary) The data for the demand of a product (in 1000's of units) over the past six-months is captured for various prices in Table-1. If the relationship between the price and demand is modelled as a simple linear regression, then answer the given sub-questions

Price of Product (in Rs.)	Demand for Product (in 1000's of units)
10	137.50
15	121.25
20	105.00
25	88.75
30	72.50
35	56.25

Table-1: Price-Demand Data

41. [Source Q2; SA]
What is the market size for the product? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 169.89 to 170.50]

42. [Source Q3; SA]
What is the satiating price for the provided data? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 52.00 to 52.50]

43. [Source Q4; SA]
At a price of Rs. 22.50, what is the elasticity? (Note: Enter your answer in decimals rounded to two decimal places. For example, if your answer is "0.245", enter it as "0.25")

[Source ans: 0.71 to 0.78]

44. [Source Q5; MCQ]
Based on the value of the elasticity at a price of Rs. 22.50, what can you conclude?

(a) It is an inelastic product demand

(b) It is an elastic product demand

(c) I don't know what it is but I want marks for this question (Note: This is not the correct answer, so don't choose it just because it suits your need :P)

[Source ans: a]

Comprehension for questions 45-46 (source Q14)

For Figure-3 given below, answer the given sub-questions

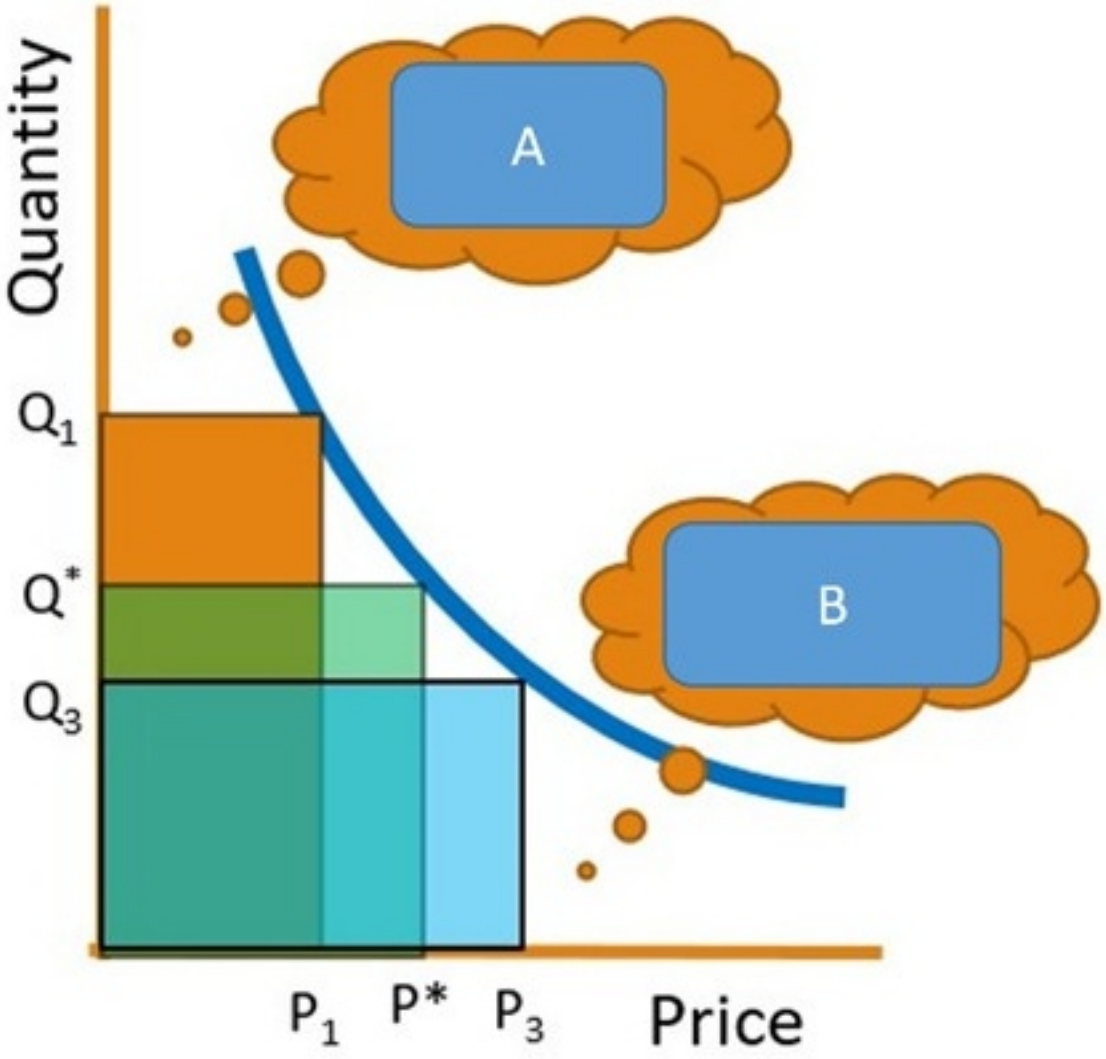


Figure-3

45. [Source Q15; MCQ]
What does region "A" represent?

(a) Market Size

(b) Latent demand

(c) Consumer Surplus

(d) None of these

[Source ans: b]

46. [Source Q16; MCQ]
What does region "B" represent?

(a) Market Size

(b) Latent demand

(c) Consumer Surplus

(d) None of these

[Source ans: c]